

1 Mössbauer Spotted: Marks 10

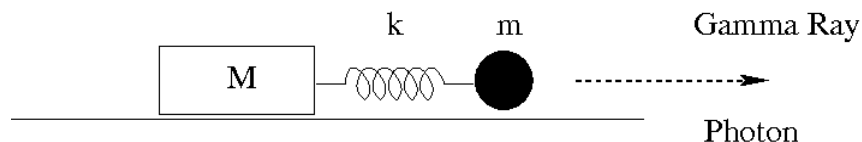
The Mössbauer Effect is a process, discovered by the German physicist Rudolf L. Mössbauer in 1958, where a nucleus emits or absorbs gamma rays without loss of energy to a nuclear recoil. It has proved to be remarkably useful for basic research in physics and chemistry, for instance, in precise measurement of small energy changes in nuclei, atoms, and crystals induced by electrical, magnetic, or gravitational fields.

Consider a “Free” nucleus of mass m with an initial energy E_i , which emits a gamma-ray photon and ends up with the final energy E_f . Initially the nucleus is at rest and after the decay let its speed be v . We assume the motion of the nucleus to be non-relativistic.

- a) Write the energy and momentum conservation laws for the decay process. [2]
 - b) After solving the above equations find the energy of the emitted gamma-ray photon in terms of $E_0 \equiv E_i - E_f$ and other quantities mentioned above. [1]
 - c) Does your answer match with the Bohr’s expression for atomic transitions? Write one or two sentences in support of your answer. (Don’t worry about the quality of your English). [1]
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Next consider the case where the nucleus is not treated as a “free” object but rather “bound inside the lattice”. We make a model of this by treating the lattice as another block of mass M which is bound to the nucleus with a spring whose constant is k . It should be apparent that $M \gg m$ but nonetheless we are not going to take it to be infinite at the moment.

We now allow the nuclear decay to take place as before. For the sake of simplicity we assume the motion of the (nucleus + Block + spring) system to take place along the same line as the momentum of the emitted gamma-ray.



- d) What would be the energy and momentum conservation equations for the “new” system? [2]
- e) Proceeding as in the earlier case, and introducing the reduced mass $\mu \equiv \frac{mM}{m+M}$ and the relative velocity between the masses find the energy of the emitted gamma-ray. [3]
- f) What is the difference with the “free” nucleus case? [1]

2 Stimulated by a LASER beam: Marks 8

LASER is an integral part of our modern day technology. It employs the concept of Stimulated radiation, which was introduced by Einstein in his derivation of the Planck formula. Following Einstein's argument, let us assume that we have a single species of atoms immersed in a bath of radiation and to simplify the calculations let us further assume that the system has two energy levels 1 and 2 with energies E_1 and E_2 respectively with $E_1 > E_2$. Let the number of atoms in levels 1 and 2 be N_1 and N_2 respectively.

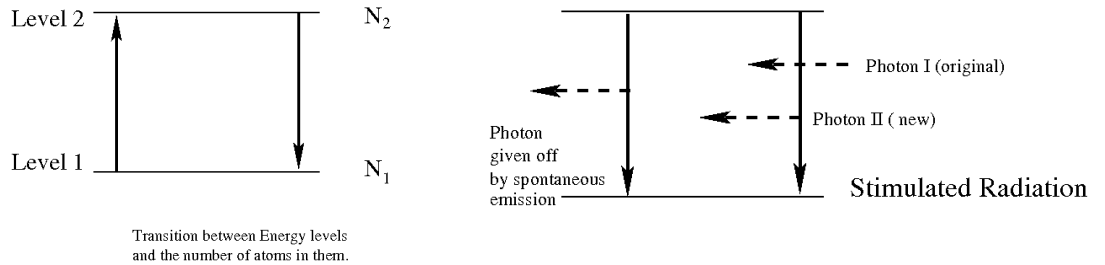
The radiation consists of photons which are absorbed as well as emitted by the atoms. Atoms in level 1 can excite to level 2 by absorbing photons while atoms in level 2 can decay to level 1 either by decaying (spontaneously). Note that the $1 \rightarrow 2$ transition thus depends on the density of the photons. So the rate at which atoms undergo the $1 \rightarrow 2$ transition is given by

$$n_{1 \rightarrow 2} = B\rho(E)N_1$$

while the spontaneous $2 \rightarrow 1$ decay will be given by

$$n'_{2 \rightarrow 1} = AN_2$$

- a) Assuming thermal equilibrium and the absence of any other process, find the ratio $\frac{A}{B}$ at a function of temperature. Assume that the atoms themselves satisfy the Boltzmann distribution. (Hint: You need to use the Boltzmann distribution $n = ge^{-\frac{E}{kT}}$, where n is the number of particles carrying energy E among a total of g particles). [2]



To reproduce the Planck formula Einstein introduced the concept of stimulated radiation where the atom decays from level 2 by emitting a photon II which is an exact copy of the already present photon I (see figure).

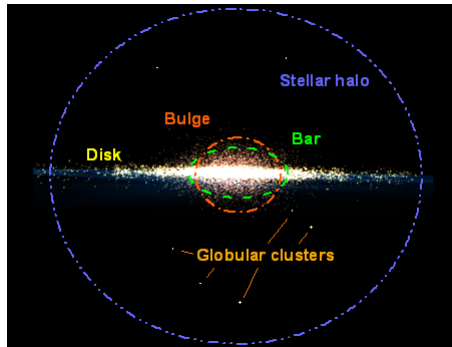
- b) What would be the rate at which the stimulated decay will take place? [1]
 c) What will be the new equilibrium condition after the introduction of this new process? [2]
 d) From the above using appropriate arguments deduce the Planck-Bose distribution for Photons. [3]

3 Where is all the Matter? : Marks 9

Vera Rubin is credited with the discovery of the galaxy rotation problem, which led to the dark matter puzzle. We will retrace the steps involved in understanding the problem and its possible resolution(s).

Let us begin with a disk shaped galaxy -the accompanying figure shows the side view (that is seen from the edge towards the center) of the galaxy. The galaxy has a bulging center which is surrounded by a disk like structure. Such galaxies are often seen to be rotating about their individual centers along the plane containing the galaxy.

- a) Consider a star belonging the disk of the galaxy which is rotating about the center. Find the equation that gives its angular frequency of the rotation as a function of the distance from the galaxy center and the mass of the galaxy that is contained within the orbit of the star. [1]



- b) Assume that the mass of the galaxy is all contained within a sphere of radius R_0 . Thus obtain the graph of the angular frequency $\omega(r)$ as a function of the orbit radius r . Your plot must cover both $r < R_0$ and $r > R_0$ regions. [2]
- c) Observation of the rotation rates of the galaxies by Vera Rubin and coworkers produced a curve of the following form for a large number of galaxies.

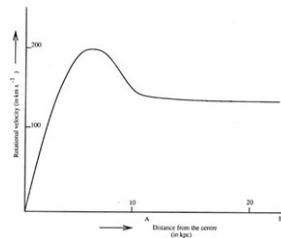


Figure 1: Angular velocity vs radial distance for an observed galaxy

How could one explain this curve?

[2]

- d) Modified Newtonian Dynamics (MOND) provides another possible explanation of this result. One claims that for very small accelerations, compared to a fixed value a_0 , the Newton's law assumes a different form.

$$m\mathbf{a}\mu\left(\frac{a_0}{a}\right) = \mathbf{F}_N$$

where $\mu(x)$ is an interpolating function, which is linear for large values of x . $\mathbf{F}_N$ is the standard Newtonian gravitational force.

- e) Show that this choice indeed is required for establishing the standard Newton's law. [1]
- f) A standard choice for the interpolating function is $\mu(x) = \frac{1}{\sqrt{1+\frac{1}{x^2}}}$. How does this choice produces the observed behavior of the rotation curves? [3]

4 Confused Experimentalist: Marks 3

- a) An experimentalist intends to measure the volume of a box and thus measures the three sides of the box with a meter stick whose smallest division is 1mm. The sides are measured 11.4mm, 23.7mm and 7.3mm.
- a) How would you write the measurement results for these three sides? [1]
- b) What would be the result for your volume measurement and the error incorporated in it? [2]